

Norms

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Properties of ℓ_∞ norm

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Introduction to norms

Introduction to norms

- ▶ A **norm** is a mathematical function that assigns a non-negative length or size to vectors.
- ▶ While there are many different types of norms, they all share three essential properties that ensure they behave consistently and meaningfully.

Definition: Norm

A function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is called a **norm** if it satisfies the following three properties for all vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ and all scalars $\lambda \in \mathbb{R}$:

1. Non-negativity and definiteness:

$$\|\mathbf{p}\| \geq 0 \quad \text{and} \quad \|\mathbf{p}\| = 0 \iff \mathbf{p} = \mathbf{0}$$

2. Absolute homogeneity (or scalability):

$$\|\lambda \mathbf{p}\| = |\lambda| \|\mathbf{p}\|$$

3. Triangle inequality (or subadditivity):

$$\|\mathbf{p} + \mathbf{w}\| \leq \|\mathbf{p}\| + \|\mathbf{w}\|$$

- ▶ **Non-negativity and definiteness** give the norm a reliable meaning as “size” or “length.” They ensure that the norm is never negative and that the only vector with zero length is the zero vector.
- ▶ **Absolute homogeneity** ensures that scaling a vector scales its length proportionally. This matches intuition: stretching a vector makes it longer in the same proportion, while reversing its direction does not change its length.
- ▶ **Triangle inequality** captures the idea that taking a detour cannot be shorter than going directly. In other words, combining two displacements cannot produce a length larger than the sum of their lengths, reflecting the “straight line is shortest” intuition behind distance.

- ▶ Different norms may emphasize different aspects of a vector (e.g., its largest component, the sum of absolute values, or the geometric length), but they all respect these fundamental axioms.

Summary

Norms:

- ▶ A **norm** $\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}$ assigns a non-negative length to every vector
- ▶ Three defining properties: **non-negativity and definiteness, absolute homogeneity, triangle inequality**
- ▶ Different norms measure different aspects of a vector, but all satisfy these axioms

Next:

- ▶ We study the ℓ_1 , ℓ_2 , generalized ℓ_2 and ℓ_∞ norms, and then we compare them

ℓ_1 norm

Introduction: ℓ_1 norm

- ▶ In this section we introduce the ℓ_1 **norm**, which measures a vector by the **sum of the absolute values** of its components
- ▶ In $\mathbb{R}^2$ it corresponds to the distance travelled along a grid of streets (**Manhattan distance**)
- ▶ We will see its geometric interpretation, its unit ball (a **diamond**) and we will prove that it satisfies the three properties of a norm

Definition 1: ℓ_1 norm

The ℓ_1 **norm** (also called **Manhattan norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j|$$

- ▶ The ℓ_1 **norm** represents the sum of the absolute values of all components.
- ▶ In $\mathbb{R}^2$, the ℓ_1 norm corresponds to the **Manhattan distance** (the distance a taxi would travel on a grid-like street network).
- ▶ The ℓ_1 norm is widely used in **robust statistics** and **sparse optimization** (e.g., Lasso regression).

Example: ℓ_1 norm

Consider the vectors:

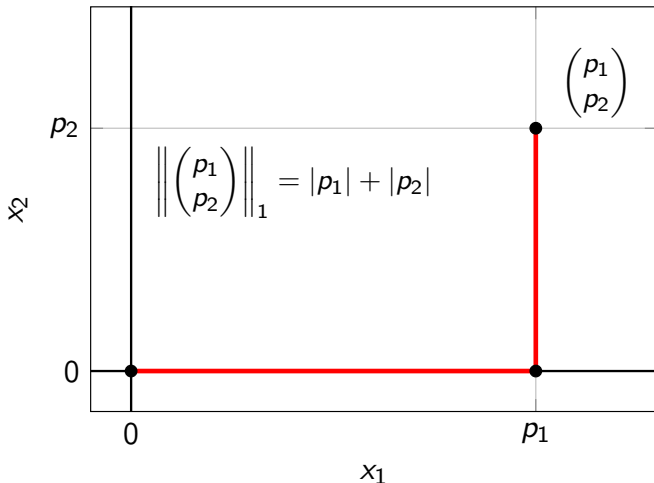
$$\mathbf{p} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} \in \mathbb{R}^2, \quad \mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \in \mathbb{R}^4$$

Their ℓ_1 norms are:

$$\left\| \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right\|_1 = |3| + |-4| = 7$$

$$\left\| \begin{pmatrix} 1 \\ 2 \\ -2 \\ 3 \end{pmatrix} \right\|_1 = |1| + |2| + |-2| + |3| = 8$$

- ▶ The ℓ_1 norm of a column vector in $\mathbb{R}^n$ represents the **sum of the absolute values of its components**.
- ▶ In $\mathbb{R}^2$, this corresponds to the **Manhattan distance** (the distance traveled along grid lines):

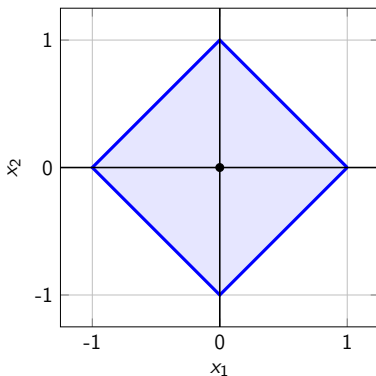


Unit ball in ℓ_1 norm

- In two dimensions, all the points at ℓ_1 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_1 \leq 1\}$$

are the points that lie on or inside a **diamond shape** (square rotated by 45°) centered at the origin.



We now show that the ℓ_1 norm satisfies the three fundamental properties that define a norm.

Non-negativity and definiteness

Observation 1: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_1 \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_1 = 0 \iff \mathbf{p} = \mathbf{0}$$

Proof: Non-negativity and definiteness

Non-negativity and definiteness.

From the definition $\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j|$.

Non-negativity: Since $|p_j| \geq 0$ for all j , and the sum of non-negative numbers is non-negative, we have $\|\mathbf{p}\|_1 \geq 0$.

Definiteness:

($\Rightarrow$) If $\|\mathbf{p}\|_1 = 0$, then $\sum_{j=1}^n |p_j| = 0$. Since each $|p_j| \geq 0$, the sum equals zero only if $|p_j| = 0$ for all j , hence $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $\|\mathbf{p}\|_1 = 0$.



Absolute homogeneity

Observation 2: Absolute homogeneity

For every scalar $\lambda \in \mathbb{R}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_1 = |\lambda| \|\mathbf{p}\|_1$$

Proof: Absolute homogeneity

Proof.

For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\lambda \mathbf{p}\|_1 &= \sum_{j=1}^n |\lambda p_j| \\ &= \sum_{j=1}^n |\lambda| |p_j| \\ &\quad \text{(property of absolute value)} \\ &= |\lambda| \sum_{j=1}^n |p_j| \\ &= |\lambda| \|\mathbf{p}\|_1\end{aligned}$$



Triangle inequality

Observation 3: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_1 \leq \|\mathbf{p}\|_1 + \|\mathbf{w}\|_1$$

Proof: Triangle inequality

Proof.

For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\mathbf{p} + \mathbf{w}\|_1 &= \sum_{j=1}^n |p_j + w_j| \\ &\leq \sum_{j=1}^n (|p_j| + |w_j|) \\ &\quad \text{(triangle inequality for abs. value)} \\ &= \sum_{j=1}^n |p_j| + \sum_{j=1}^n |w_j| \\ &= \|\mathbf{p}\|_1 + \|\mathbf{w}\|_1\end{aligned}$$



Summary

ℓ_1 norm (Manhattan norm):

- ▶ $\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j|$: sum of the absolute values of the components
- ▶ In $\mathbb{R}^2$: Manhattan distance (distance travelled along grid lines)
- ▶ Unit ball in $\mathbb{R}^2$: **diamond** with vertices $(\pm 1, 0)$ and $(0, \pm 1)$
- ▶ It satisfies non-negativity and definiteness, absolute homogeneity and the triangle inequality

ℓ_2 norm

Introduction: ℓ_2 norm

- ▶ In this section we introduce the ℓ_2 **norm (Euclidean norm)**, the most common way of measuring the length of a vector
- ▶ It is the usual distance from the origin and follows from the **Pythagorean theorem**
- ▶ We will see the norm of the sum and of the difference of two vectors, the **Cauchy–Schwarz inequality**, the properties of a norm and the (reverse) **triangle inequality**

Definition 2: ℓ_2 norm

The ℓ_2 **norm** (also called **Euclidean norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_2 = \underbrace{\sqrt{\sum_{j=1}^n p_j^2}}_{\sqrt{\mathbf{p}' \mathbf{p}}}$$

- ▶ When the subscript is omitted, i.e., when we write $\|\mathbf{p}\|$ instead of $\|\mathbf{p}\|_2$, the ℓ_2 norm is implied by default.

Example: ℓ_2 norm (1/2)

- Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$$

Their norms are:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17}$$

$$\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{4^2 + 1^2} = \sqrt{16 + 1} = \sqrt{17}$$

Example: ℓ_2 norm (2/2)

- Consider the four vertices of the diamond-shaped ℓ_1 unit ball in $\mathbb{R}^2$:

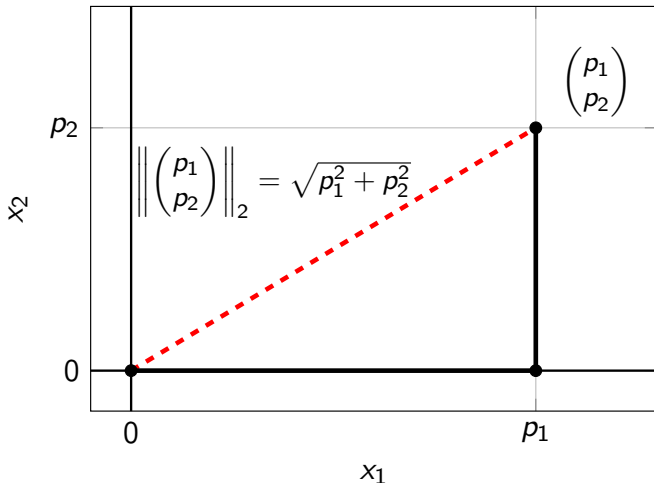
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Their norms are:

$$\left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|_2 = \sqrt{1^2 + 0^2} = 1, \quad \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{0^2 + 1^2} = 1$$

$$\left\| \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\|_2 = \sqrt{(-1)^2 + 0^2} = 1, \quad \left\| \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\|_2 = \sqrt{0^2 + (-1)^2} = 1$$

- ▶ The ℓ_2 norm of a column vector in $\mathbb{R}^n$ represents its **Euclidean distance from the origin**.
- ▶ This interpretation follows from the **Pythagorean theorem**, as illustrated in $\mathbb{R}^2$ by the following picture:

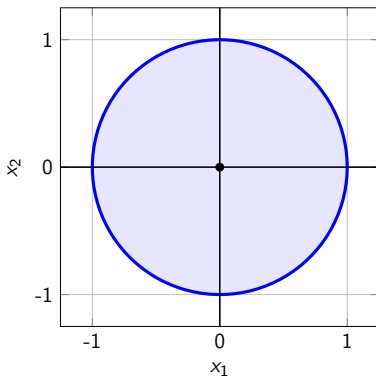


Unit ball in ℓ_2 norm

- In two dimensions, all the points at ℓ_2 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_2 \leq 1\}$$

are the points that lie on or inside the **unit circle** centered at the origin.



Norm of the difference

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the norm of the **difference** of the two vectors is:

$$\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{\sum_{j=1}^n (p_j - w_j)^2}$$

This quantity corresponds both to the **Euclidean distance** between the two points $\mathbf{p}$ and $\mathbf{w}$, and to the **length of the diagonal** of the parallelogram spanned by those points, starting from one and pointing to the other.

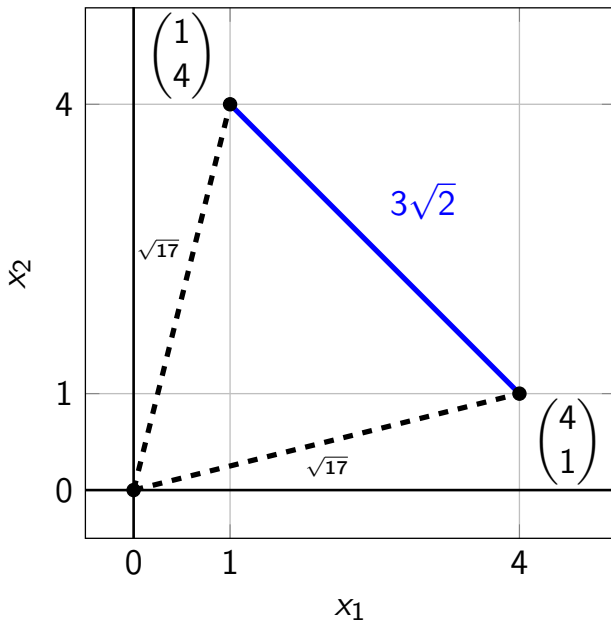
Example: ℓ_2 norm of the difference

Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$$

The norm of the difference of the two vectors is

$$\begin{aligned} \left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 &= \sqrt{(1-4)^2 + (4-1)^2} \\ &= \sqrt{(-3)^2 + 3^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \end{aligned}$$



Norm of the sum

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the norm of the **sum** of the two vectors is:

$$\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{\sum_{j=1}^n (p_j + w_j)^2}$$

This quantity represents the **length of the diagonal** of the parallelogram spanned by the vectors $\mathbf{p}$ and $\mathbf{w}$, both originating from the origin.

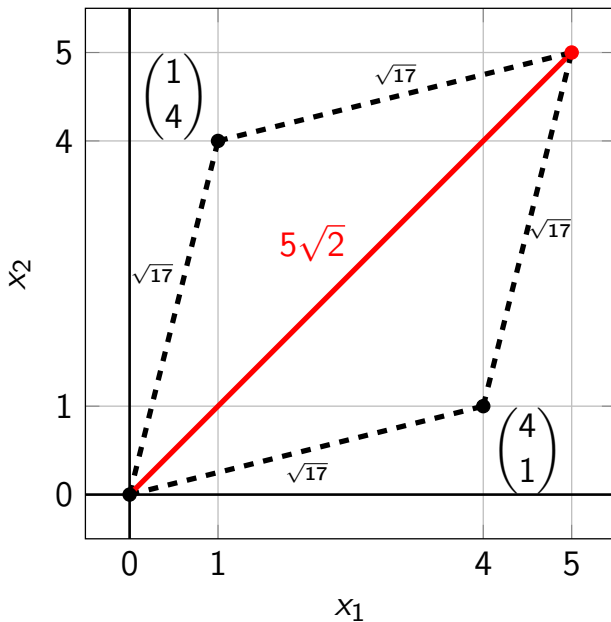
Example: ℓ_2 norm of the sum

Consider the two vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2 \quad \text{and} \quad \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$$

The norm of the sum of the two vectors is:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$



Cauchy–Schwarz inequality

Observation 4: Cauchy–Schwarz inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$|\mathbf{p}' \mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$$

Proof of Cauchy–Schwarz inequality (1/3)

Proof of Cauchy–Schwarz inequality (1/3).

If $\mathbf{p} = 0$ or $\mathbf{w} = 0$ (or both), the inequality clearly holds.

Assume $\mathbf{p} \neq 0$ and $\mathbf{w} \neq 0$. Consider the vector:

$$\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}, \quad \text{with } \alpha, \beta \in \mathbb{R}$$



Proof of Cauchy–Schwarz inequality (2/3)

Proof of Cauchy–Schwarz inequality (2/3).

For every $\alpha, \beta \in \mathbb{R}$, we have:

$$\underbrace{\mathbf{u}' \mathbf{u}}_{\geq 0} = (\alpha \mathbf{p} + \beta \mathbf{w})' (\alpha \mathbf{p} + \beta \mathbf{w}) \\ = \alpha^2 \mathbf{p}' \mathbf{p} + 2\alpha\beta \mathbf{p}' \mathbf{w} + \beta^2 \mathbf{w}' \mathbf{w} \geq 0$$

Now choose:

$$\alpha = \mathbf{w}' \mathbf{w}, \quad \beta = -\mathbf{p}' \mathbf{w}$$



Proof of Cauchy–Schwarz inequality (3/3)

Proof of Cauchy–Schwarz inequality (3/3).

Substituting, we get:

$$\begin{aligned} & (\mathbf{w}' \mathbf{w})^2 \mathbf{p}' \mathbf{p} - 2 (\mathbf{p}' \mathbf{w})^2 (\mathbf{w}' \mathbf{w}) \\ & \quad + (\mathbf{p}' \mathbf{w})^2 (\mathbf{w}' \mathbf{w}) \\ & = \mathbf{w}' \mathbf{w} ((\mathbf{w}' \mathbf{w}) \mathbf{p}' \mathbf{p} - (\mathbf{p}' \mathbf{w})^2) \geq 0 \end{aligned}$$

Since $\mathbf{w}' \mathbf{w} = \|\mathbf{w}\|_2^2 > 0$, dividing both sides yields:

$$\begin{aligned} & (\mathbf{p}' \mathbf{w})^2 \leq \mathbf{p}' \mathbf{p} \mathbf{w}' \mathbf{w} \\ \implies & |\mathbf{p}' \mathbf{w}| \leq \sqrt{\mathbf{p}' \mathbf{p}} \sqrt{\mathbf{w}' \mathbf{w}} = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \end{aligned}$$



We now show that the ℓ_2 norm satisfies the three fundamental properties that define a norm.

Non-negativity and definiteness

Observation 5: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_2 \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_2 = 0 \iff \mathbf{p} = \mathbf{0}$$

Proof: Non-negativity and definiteness

Non-negativity and definiteness.

From the definition $\|\mathbf{p}\|_2 = \sqrt{\mathbf{p}'\mathbf{p}}$.

Non-negativity: Since $\mathbf{p}'\mathbf{p} = \sum_{j=1}^n p_j^2 \geq 0$, we have $\|\mathbf{p}\|_2 \geq 0$.

Definiteness:

($\Rightarrow$) If $\|\mathbf{p}\|_2 = 0$, then $\sum_{j=1}^n p_j^2 = 0$. Since each $p_j^2 \geq 0$, the sum equals zero only if $p_j = 0$ for all j , i.e., $\mathbf{p} = \mathbf{0}$.

($\Leftarrow$) If $\mathbf{p} = \mathbf{0}$, then $\|\mathbf{p}\|_2 = 0$.



Absolute homogeneity

Observation 6: Absolute homogeneity

For every column vector $\mathbf{p} \in \mathbb{R}^n$ and scalar $\lambda \in \mathbb{R}$, we have:

$$\|\lambda \mathbf{p}\|_2 = |\lambda| \|\mathbf{p}\|_2$$

Proof: Absolute homogeneity

Absolute homogeneity (ℓ_2 norm).

For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\lambda \mathbf{p}\|_2 &= \sqrt{\sum_{j=1}^n (\lambda p_j)^2} \\ &= \sqrt{\sum_{j=1}^n \lambda^2 p_j^2} \\ &= |\lambda| \sqrt{\sum_{j=1}^n p_j^2} \\ &= |\lambda| \|\mathbf{p}\|_2\end{aligned}$$



Triangle inequality

Observation 7: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2$$

Proof: Triangle inequality

Triangle inequality (ℓ_2 norm).

For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\mathbf{p} + \mathbf{w}\|_2^2 &= (\mathbf{p} + \mathbf{w})' (\mathbf{p} + \mathbf{w}) \\ &= \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2 \mathbf{p}' \mathbf{w} \\ &\leq \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2 \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \\ &\quad \text{(by Cauchy-Schwarz)} \\ &= (\|\mathbf{p}\|_2 + \|\mathbf{w}\|_2)^2\end{aligned}$$

Accordingly:

$$\|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2$$



- ▶ In $\mathbb{R}^2$, the triangle inequality states that **for any triangle the sum of the lengths of any two sides must be greater than or equal to the length of the remaining side.**

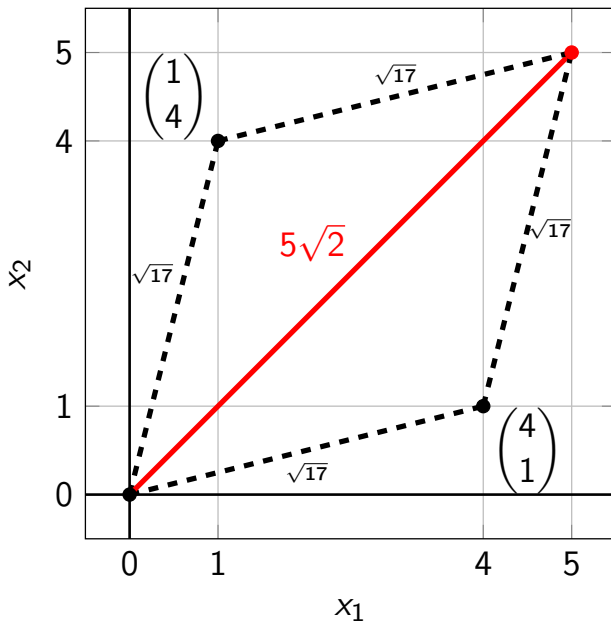
Example: Triangle inequality (1/2)

Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$, we have:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{17}, \quad \left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{17}$$

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = 5\sqrt{2}$$

$$5\sqrt{2} \leq \sqrt{17} + \sqrt{17}$$



- ▶ Another form of the triangle inequality is:

$$\|\mathbf{r}\|_2 \leq \|\mathbf{r} - \mathbf{s}\|_2 + \|\mathbf{s}\|_2$$

that is

$$\|\mathbf{r}\|_2 - \|\mathbf{s}\|_2 \leq \|\mathbf{r} - \mathbf{s}\|_2, \quad \forall \mathbf{r}, \mathbf{s} \in \mathbb{R}^n$$

- ▶ To obtain it, it suffices to set in the triangle inequality:

$$\mathbf{p} = \mathbf{r} - \mathbf{s}, \quad \mathbf{w} = \mathbf{s}$$

Reverse triangle inequality

Observation 8: reverse triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$| \|\mathbf{p}\|_2 - \|\mathbf{w}\|_2 | \leq \|\mathbf{p} - \mathbf{w}\|_2$$

Proof: Reverse triangle inequality

Reverse triangle inequality.

For every two vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\mathbf{p} - \mathbf{w}\|_2^2 &= (\mathbf{p} - \mathbf{w})' (\mathbf{p} - \mathbf{w}) \\ &= \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \mathbf{p}' \mathbf{w} \\ &\geq \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 - 2 \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 \\ &\quad \text{(by Cauchy-Schwarz)} \\ &= (\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2)^2\end{aligned}$$

Accordingly:

$$|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2$$



- ▶ In $\mathbb{R}^2$, the reverse triangle inequality states that **for any triangle the length of any side must be greater than or equal to the difference between the lengths of the other two sides.**

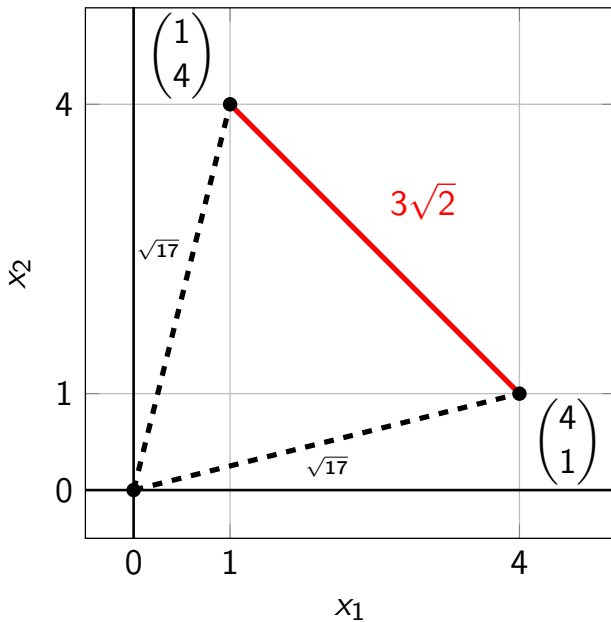
Example: Reverse triangle inequality (1/2)

Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$, we have:

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\|_2 = \sqrt{17}, \quad \left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = \sqrt{17}$$

$$\left\| \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|_2 = 3\sqrt{2}$$

$$\left| \sqrt{17} - \sqrt{17} \right| \leq 3\sqrt{2}$$



Summary (1/2)

ℓ_2 norm (Euclidean norm):

- ▶ $\|\mathbf{p}\|_2 = \sqrt{\sum_{j=1}^n p_j^2} = \sqrt{\mathbf{p}' \mathbf{p}}$: Euclidean distance from the origin (Pythagorean theorem)
- ▶ Unit ball in $\mathbb{R}^2$: the **unit circle**
- ▶ $\|\mathbf{p} - \mathbf{w}\|_2$ is the distance between $\mathbf{p}$ and $\mathbf{w}$; $\|\mathbf{p} + \mathbf{w}\|_2$ is the length of the diagonal of the parallelogram

Summary (2/2)

Inequalities:

- ▶ Cauchy–Schwarz: $|\mathbf{p}' \mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$
- ▶ Triangle inequality: $\|\mathbf{p} + \mathbf{w}\|_2 \leq \|\mathbf{p}\|_2 + \|\mathbf{w}\|_2$ (proved via Cauchy–Schwarz)
- ▶ Reverse triangle inequality: $|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2$

Properties:

- ▶ The ℓ_2 norm satisfies non-negativity and definiteness, absolute homogeneity and the triangle inequality

Generalized ℓ_2 norm

Introduction: Generalized ℓ_2 norm

- ▶ In this section we extend the ℓ_2 norm by means of a **symmetric positive definite matrix Q** , which weights and potentially correlates the components of the vector
- ▶ In $\mathbb{R}^2$, the unit ball becomes an **ellipse**, whose shape and orientation depend on the eigenvalues and eigenvectors of Q
- ▶ We will see three examples, the **generalized Cauchy–Schwarz inequality** and the three properties of a norm

Generalized ℓ_2 norm

Definition 3: generalized ℓ_2 norm

Given a symmetric positive definite matrix

$$\underbrace{\begin{pmatrix} q_{11} & q_{12} & \dots & q_{1n} \\ q_{21} & q_{22} & \dots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{n1} & q_{n2} & \dots & q_{nn} \end{pmatrix}}_Q \in \mathbb{R}^{n \times n}$$

the **generalized ℓ_2 norm** of a column vector $\mathbf{p} \in \mathbb{R}^n$ is:

$$\|\mathbf{p}\|_Q = \underbrace{\sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} p_i p_j}}_{\sqrt{\mathbf{p}' Q \mathbf{p}}}$$

Properties of generalized ℓ_2 norm

- ▶ The **generalized ℓ_2 norm** extends the standard Euclidean norm by incorporating a positive definite matrix $\mathbf{Q}$, which weights and potentially correlates the components of the vector.
- ▶ When $\mathbf{Q}$ is the identity matrix $\mathbf{I}$, the generalized ℓ_2 norm reduces to the standard ℓ_2 norm.
- ▶ The matrix $\mathbf{Q}$ must be **positive definite** to ensure that $\mathbf{p}'\mathbf{Q}\mathbf{p} \geq 0$ for all $\mathbf{p} \in \mathbb{R}^n$, making the square root well-defined, and that $\mathbf{p}'\mathbf{Q}\mathbf{p} = 0$ only for $\mathbf{p} = 0$.

Unit ball in generalized ℓ_2 norm

- ▶ In two dimensions, all the points at generalized ℓ_2 norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_Q \leq 1\}$$

are the points that lie on or inside an **ellipse** centered at the origin, whose size, shape, and rotation are determined by the matrix Q .

- ▶ In particular, the eigenvectors of Q determine the principal axes and the orientation, while the eigenvalues determine the lengths of the semi-axes.

Example: Generalized ℓ_2 norm (1/3)

In $\mathbb{R}^2$, we consider the following diagonal positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

Since $\mathbf{Q}$ is **diagonal**, the eigenvalues are the diagonal entries:

$$\lambda_1 = 4, \quad \lambda_2 = 1$$

and the eigenvectors are the standard basis vectors:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{for } \lambda_1 = 4), \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_2 = 1)$$

Example: Generalized ℓ_2 norm (2/3)

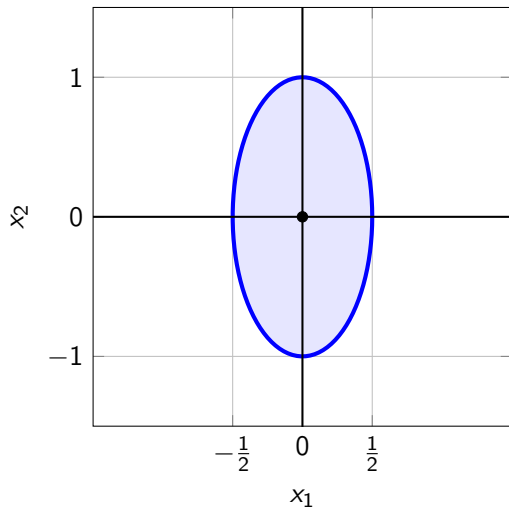
Semi-axes lengths and orientation:

- ▶ The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{1}} = 1$.
- ▶ Since the eigenvectors are aligned with the coordinate axes, the ellipse is **not rotated**.
- ▶ The principal axes coincide with the x_1 and x_2 axes.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\begin{aligned}\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_Q \leq 1\} &= \left\{ \mathbf{x} \in \mathbb{R}^2 : \sqrt{4x_1^2 + x_2^2} \leq 1 \right\} \\ &= \left\{ \mathbf{x} \in \mathbb{R}^2 : \frac{x_1^2}{(1/2)^2} + \frac{x_2^2}{1^2} \leq 1 \right\}\end{aligned}$$

Example: Generalized ℓ_2 norm (3/3)



Example: Generalized ℓ_2 norm (1/3)

In $\mathbb{R}^2$, we consider the following diagonal positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} \frac{1}{9} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

Since $\mathbf{Q}$ is **diagonal**, the eigenvalues are the diagonal entries:

$$\lambda_1 = \frac{1}{9}, \quad \lambda_2 = \frac{1}{4}$$

and the eigenvectors are the standard basis vectors:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{for } \lambda_1 = \frac{1}{9}), \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_2 = \frac{1}{4})$$

Example: Generalized ℓ_2 norm (2/3)

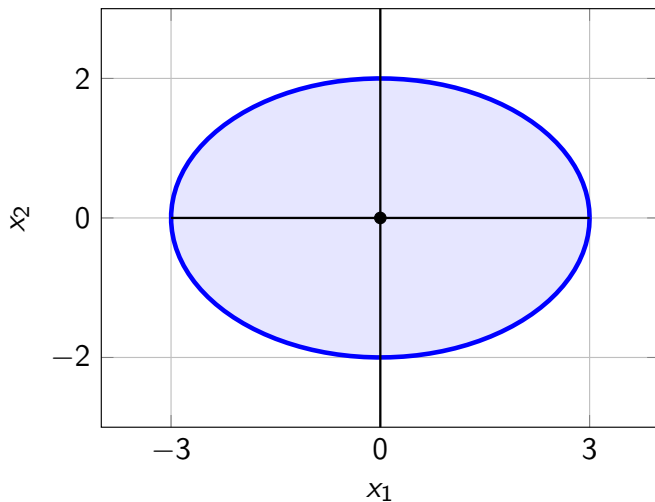
Semi-axes lengths and orientation:

- ▶ The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{1/9}} = 3$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{1/4}} = 2$.
- ▶ Since the eigenvectors are aligned with the coordinate axes, the ellipse is **not rotated**.
- ▶ The principal axes coincide with the x_1 and x_2 axes.

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\begin{aligned}\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} &= \left\{ \mathbf{x} \in \mathbb{R}^2 : \sqrt{\frac{x_1^2}{9} + \frac{x_2^2}{4}} \leq 1 \right\} \\ &= \left\{ \mathbf{x} \in \mathbb{R}^2 : \frac{x_1^2}{3^2} + \frac{x_2^2}{2^2} \leq 1 \right\}\end{aligned}$$

Example: Generalized ℓ_2 norm (3/3)



Example: Generalized ℓ_2 norm (1/4)

In $\mathbb{R}^2$, we consider the following symmetric positive definite matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The eigenvalues of $\mathbf{Q}$ are found by solving the characteristic equation:

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & -1 \\ -1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$$

which gives:

$$\lambda_1 = 1, \quad \lambda_2 = 3$$

Example: Generalized ℓ_2 norm (2/4)

The corresponding eigenvectors are:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (\text{for } \lambda_1 = 1), \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (\text{for } \lambda_2 = 3)$$

Semi-axes lengths and orientation:

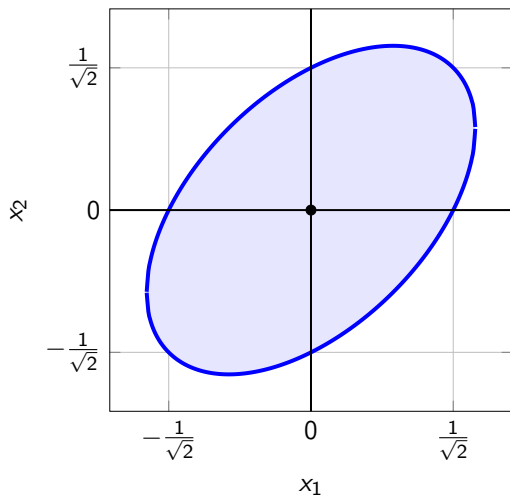
- ▶ The semi-axes lengths are $\frac{1}{\sqrt{\lambda_1}} = \frac{1}{\sqrt{1}} = 1$ and $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{3}}$.
- ▶ Since the eigenvectors are **not aligned** with the coordinate axes, the ellipse is **rotated**.
- ▶ The principal axes coincide with the directions of the eigenvectors.

Example: Generalized ℓ_2 norm (3/4)

The set of points at generalized ℓ_2 norm ≤ 1 from the origin is:

$$\begin{aligned} & \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_{\mathbf{Q}} \leq 1\} \\ &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : \sqrt{2x_1^2 - 2x_1x_2 + 2x_2^2} \leq 1 \right\} \\ &= \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : 2x_1^2 - 2x_1x_2 + 2x_2^2 \leq 1 \right\} \end{aligned}$$

Example: Generalized ℓ_2 norm (4/4)



Generalized Cauchy–Schwarz inequality

Observation 9: generalized Cauchy–Schwarz inequality

For every symmetric positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$|\mathbf{p}' \mathbf{Q} \mathbf{w}| \leq \|\mathbf{p}\|_{\mathbf{Q}} \|\mathbf{w}\|_{\mathbf{Q}}$$

Proof: Generalized Cauchy–Schwarz inequality (1/3)

Generalized Cauchy–Schwarz inequality (1/3).

If $\mathbf{p} = 0$ or $\mathbf{w} = 0$ (or both), the inequality clearly holds.

Assume $\mathbf{p} \neq 0$ and $\mathbf{w} \neq 0$. Consider the vector:

$$\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}, \quad \text{with } \alpha, \beta \in \mathbb{R}$$



Proof: Generalized Cauchy–Schwarz inequality (2/3)

Generalized Cauchy–Schwarz inequality (2/3).

Then, for every $\alpha, \beta \in \mathbb{R}$, we have:

$$\begin{aligned}\underbrace{\mathbf{u}' \mathbf{Q} \mathbf{u}}_{\geq 0} &= (\alpha \mathbf{p} + \beta \mathbf{w})' \mathbf{Q} (\alpha \mathbf{p} + \beta \mathbf{w}) \\ &= \alpha^2 \mathbf{p}' \mathbf{Q} \mathbf{p} + 2\alpha\beta \mathbf{p}' \mathbf{Q} \mathbf{w} \\ &\quad + \beta^2 \mathbf{w}' \mathbf{Q} \mathbf{w} \geq 0\end{aligned}$$

where we used $\mathbf{w}' \mathbf{Q} \mathbf{p} = \mathbf{p}' \mathbf{Q} \mathbf{w}$, which holds since $\mathbf{Q}$ is symmetric.

Now choose:

$$\alpha = \mathbf{w}' \mathbf{Q} \mathbf{w}, \quad \beta = -\mathbf{p}' \mathbf{Q} \mathbf{w}$$



Proof: Generalized Cauchy–Schwarz inequality (3/3)

Generalized Cauchy–Schwarz inequality (3/3).

Substituting yields:

$$\mathbf{w}' \mathbf{Q} \mathbf{w} ((\mathbf{w}' \mathbf{Q} \mathbf{w}) \mathbf{p}' \mathbf{Q} \mathbf{p} - (\mathbf{p}' \mathbf{Q} \mathbf{w})^2) \geq 0$$

Since $\mathbf{w}' \mathbf{Q} \mathbf{w} = \|\mathbf{w}\|_Q^2 > 0$, dividing both sides yields:

$$(\mathbf{p}' \mathbf{Q} \mathbf{w})^2 \leq \mathbf{p}' \mathbf{Q} \mathbf{p} \cdot \mathbf{w}' \mathbf{Q} \mathbf{w}$$

$$\implies |\mathbf{p}' \mathbf{Q} \mathbf{w}| \leq \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}} \sqrt{\mathbf{w}' \mathbf{Q} \mathbf{w}} = \|\mathbf{p}\|_Q \|\mathbf{w}\|_Q$$



We now show that the generalized ℓ_2 norm satisfies the three fundamental properties that define a norm.

Non-negativity and definiteness

Observation 10: Non-negativity and definiteness

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_{\mathbf{Q}} \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_{\mathbf{Q}} = 0 \iff \mathbf{p} = \mathbf{0}$$

Proof: Non-negativity and definiteness

Non-negativity and definiteness.

By definition, $\|\mathbf{p}\|_Q = \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}}$. Since $\mathbf{Q}$ is positive definite, we have $\mathbf{p}' \mathbf{Q} \mathbf{p} > 0$ for all $\mathbf{p} \neq 0$, and $0' \mathbf{Q} 0 = 0$.

Non-negativity: $\|\mathbf{p}\|_Q = \sqrt{\mathbf{p}' \mathbf{Q} \mathbf{p}} \geq 0$.

Definiteness:

($\Rightarrow$) If $\|\mathbf{p}\|_Q = 0$, then $\mathbf{p}' \mathbf{Q} \mathbf{p} = 0$, which implies $\mathbf{p} = 0$ (by positive definiteness).

($\Leftarrow$) If $\mathbf{p} = 0$, then $\|\mathbf{p}\|_Q = 0$.



Absolute homogeneity

Observation 11: Absolute homogeneity

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$, scalar $\lambda \in \mathbb{R}$, and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_{\mathbf{Q}} = |\lambda| \|\mathbf{p}\|_{\mathbf{Q}}$$

Proof: Absolute homogeneity

Absolute homogeneity.

For every scalar $\lambda \in \mathbb{R}$ and every vector $\mathbf{p} \in \mathbb{R}^n$:

$$\begin{aligned}\|\lambda \mathbf{p}\|_Q &= \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} \lambda p_i \lambda p_j} \\&= \sqrt{\sum_{i=1}^n \sum_{j=1}^n \lambda^2 q_{ij} p_i p_j} \\&= |\lambda| \sqrt{\sum_{i=1}^n \sum_{j=1}^n q_{ij} p_i p_j} \\&= |\lambda| \|\mathbf{p}\|_Q\end{aligned}$$



Triangle inequality

Observation 12: Triangle inequality

For every positive definite matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and pairs of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_{\mathbf{Q}} \leq \|\mathbf{p}\|_{\mathbf{Q}} + \|\mathbf{w}\|_{\mathbf{Q}}$$

Proof: Triangle inequality

Triangle inequality.

For every $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\mathbf{p} + \mathbf{w}\|_Q^2 &= (\mathbf{p} + \mathbf{w})' Q (\mathbf{p} + \mathbf{w}) \\&= \|\mathbf{p}\|_Q^2 + \|\mathbf{w}\|_Q^2 + 2 \mathbf{p}' Q \mathbf{w} \\&\leq \|\mathbf{p}\|_Q^2 + \|\mathbf{w}\|_Q^2 + 2 \|\mathbf{p}\|_Q \|\mathbf{w}\|_Q \\&\quad \text{(by gen. Cauchy-Schwarz)} \\&= (\|\mathbf{p}\|_Q + \|\mathbf{w}\|_Q)^2\end{aligned}$$

Accordingly:

$$\|\mathbf{p} + \mathbf{w}\|_Q \leq \|\mathbf{p}\|_Q + \|\mathbf{w}\|_Q$$



Generalized ℓ_2 norm:

- ▶ $\|\mathbf{p}\|_Q = \sqrt{\mathbf{p}'\mathbf{Q}\mathbf{p}}$, with $\mathbf{Q}$ symmetric positive definite
- ▶ It reduces to the standard ℓ_2 norm when $\mathbf{Q} = \mathbf{I}$
- ▶ Unit ball in $\mathbb{R}^2$: an **ellipse**; eigenvectors of $\mathbf{Q}$ give the principal axes, eigenvalues λ_i give the semi-axes $1/\sqrt{\lambda_i}$
- ▶ Generalized Cauchy–Schwarz: $|\mathbf{p}'\mathbf{Q}\mathbf{w}| \leq \|\mathbf{p}\|_Q \|\mathbf{w}\|_Q$
- ▶ It satisfies the three properties of a norm

l_∞ norm

Introduction: ℓ_∞ norm

- ▶ In this section we introduce the ℓ_∞ **norm**, which measures a vector by its **largest component in absolute value**
- ▶ It is useful whenever we care about the **worst case**, i.e., the largest deviation among all components
- ▶ We will see its geometric interpretation, its unit ball (a **square**) and we will prove that it satisfies the three properties of a norm

Definition 4: ℓ_∞ norm

The ℓ_∞ **norm** (also called **maximum norm**, **supremum norm**, or **Chebyshev norm**) of a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

is the non-negative number defined as:

$$\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\}$$

- ▶ The ℓ_∞ **norm** represents the maximum absolute value among all components.
- ▶ The ℓ_∞ norm is half the side length of the smallest hypercube (aligned with the axes and centered at the origin) that contains the vector.

Example: ℓ_∞ norm

Consider the vectors:

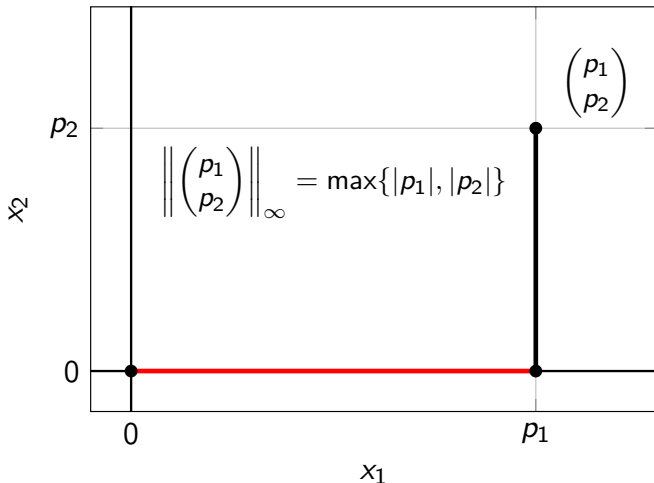
$$\mathbf{p} = \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \in \mathbb{R}^3, \quad \mathbf{w} = \begin{pmatrix} -1 \\ 5 \\ -5 \\ 2 \end{pmatrix} \in \mathbb{R}^4$$

Their ℓ_∞ norms are:

$$\left\| \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \right\|_\infty = \max\{|3|, |-7|, |2|\} = \max\{3, 7, 2\} = 7$$

$$\left\| \begin{pmatrix} -1 \\ 5 \\ -5 \\ 2 \end{pmatrix} \right\|_\infty = \max\{|-1|, |5|, |-5|, |2|\} = \max\{1, 5, 5, 2\} = 5$$

- ▶ The ℓ_∞ **norm** of a column vector in $\mathbb{R}^n$ represents the **maximum absolute value of its components**.
- ▶ In $\mathbb{R}^2$, this corresponds to the **Chebyshev distance** (the maximum of the absolute differences in coordinates):

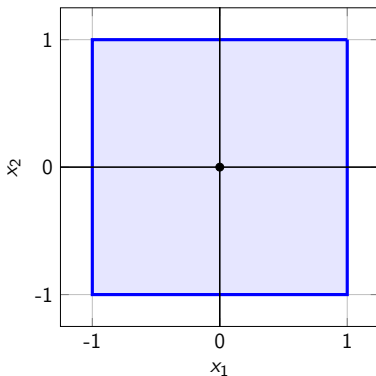


Unit ball in ℓ_∞ norm

- In two dimensions, all the points at ℓ_∞ norm less than or equal to 1

$$\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_\infty \leq 1\}$$

are the points that lie on or inside a **square** (axis-aligned) centered at the origin.



We now show that the ℓ_∞ norm satisfies the three fundamental properties that define a norm.

Non-negativity and definiteness

Observation 13: Non-negativity and definiteness

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_{\infty} \geq 0 \quad \text{and} \quad \|\mathbf{p}\|_{\infty} = 0 \iff \mathbf{p} = \mathbf{0}$$

Proof: Non-negativity and definiteness (1/2)

Non-negativity and definiteness (1/2).

From the definition $\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, 2, \dots, n\}\}$.

Non-negativity: Since $|p_j| \geq 0$ for all j , and the maximum of non-negative numbers is non-negative, we have

$$\|\mathbf{p}\|_\infty \geq 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^n$$

Definiteness ($\Rightarrow$): If $\|\mathbf{p}\|_\infty = 0$, then

$$\max\{|p_j| : j \in \{1, 2, \dots, n\}\} = 0$$



Proof: Non-negativity and definiteness (2/2)

Non-negativity and definiteness (2/2).

Since

$$|p_j| \leq \max\{|p_k| : k \in \{1, 2, \dots, n\}\}$$

for all j , it follows that

$$|p_j| \leq 0 \text{ for all } j.$$

But $|p_j| \geq 0$ always holds, hence $|p_j| = 0$ for all j , i.e. $\mathbf{p} = 0$.

($\Leftarrow$) If $\mathbf{p} = 0$, then $p_j = 0$ for all j , so

$$\|\mathbf{p}\|_\infty = \max\{|0| : j \in \{1, 2, \dots, n\}\} = 0$$



Absolute homogeneity

Observation 14: Absolute homogeneity

For every scalar $\lambda \in \mathbb{R}$ and column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\lambda \mathbf{p}\|_{\infty} = |\lambda| \|\mathbf{p}\|_{\infty}$$

Proof: Absolute homogeneity

Proof.

For every scalar $\lambda \in \mathbb{R}$ and every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\lambda \mathbf{p}\|_{\infty} &= \max\{|\lambda p_j| : j \in \{1, 2, \dots, n\}\} \\ &= \max\{|\lambda| |p_j| : j \in \{1, 2, \dots, n\}\} \\ &\quad \text{(property of absolute value)} \\ &= |\lambda| \max\{|p_j| : j \in \{1, 2, \dots, n\}\} \\ &\quad \text{(factor constant out of max)} \\ &= |\lambda| \|\mathbf{p}\|_{\infty}\end{aligned}$$



Triangle inequality

Observation 15: Triangle inequality

For every pair of column vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p} + \mathbf{w}\|_{\infty} \leq \|\mathbf{p}\|_{\infty} + \|\mathbf{w}\|_{\infty}$$

Proof: Triangle inequality

Proof.

For every pair of vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$, we have:

$$\begin{aligned}\|\mathbf{p} + \mathbf{w}\|_{\infty} &= \max\{|p_j + w_j| : j \in \{1, 2, \dots, n\}\} \\ &\leq \max\{|p_j| + |w_j| : j \in \{1, 2, \dots, n\}\} \\ &\quad \text{(triangle inequality for abs. value)} \\ &\leq \max\{|p_j| : j\} + \max\{|w_j| : j\} \\ &\quad \text{(max distribution)} \\ &= \|\mathbf{p}\|_{\infty} + \|\mathbf{w}\|_{\infty}\end{aligned}$$

where we used that $|p_j| \leq \max\{|p_k| : k\}$ and $|w_j| \leq \max\{|w_k| : k\}$ for all j . □

Summary

ℓ_∞ norm (maximum/Chebyshev norm):

- ▶ $\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, \dots, n\}\}$: largest absolute value among the components
- ▶ It is half the side length of the smallest axis-aligned hypercube centered at the origin containing the vector
- ▶ Unit ball in $\mathbb{R}^2$: the axis-aligned **square** with vertices $(\pm 1, \pm 1)$
- ▶ It satisfies non-negativity and definiteness, absolute homogeneity and the triangle inequality

Comparison of the ℓ_1 , ℓ_2 and ℓ_∞ norms

Introduction: Comparison of the norms

- ▶ The ℓ_1 , ℓ_2 and ℓ_∞ norms of the same vector are, in general, **different numbers**
- ▶ However, they are related by **simple inequalities**: if one of them is small, the others are small too
- ▶ We will prove these inequalities, check them on an example and interpret them geometrically: the three unit balls are **nested**

Inequalities between norms

Observation 16: Inequalities between norms

For every column vector $\mathbf{p} \in \mathbb{R}^n$, we have:

$$\|\mathbf{p}\|_{\infty} \leq \|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_{\infty}$$

and

$$\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2$$

Proof: Inequalities between norms (1/4)

Inequalities between norms (1/4).

Let $k \in \{1, 2, \dots, n\}$ be an index such that $|p_k| = \|\mathbf{p}\|_\infty$.

1. $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2$: since all the terms p_j^2 are non-negative, we have

$$\|\mathbf{p}\|_\infty^2 = p_k^2 \leq \sum_{j=1}^n p_j^2 = \|\mathbf{p}\|_2^2$$

and, taking the square root of both (non-negative) sides,

$$\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2$$



Proof: Inequalities between norms (2/4)

Inequalities between norms (2/4).

2. $\|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1$: expanding the square of the sum, we have

$$\begin{aligned}\|\mathbf{p}\|_1^2 &= \left(\sum_{j=1}^n |p_j| \right)^2 = \sum_{j=1}^n |p_j|^2 + \underbrace{\sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n |p_i| |p_j|}_{\geq 0} \\ &\geq \sum_{j=1}^n p_j^2 = \|\mathbf{p}\|_2^2\end{aligned}$$

and, taking the square root of both (non-negative) sides,

$$\|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1$$



Proof: Inequalities between norms (3/4)

Inequalities between norms (3/4).

3. $\|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty$: since $|p_j| \leq \|\mathbf{p}\|_\infty$ for all $j \in \{1, 2, \dots, n\}$, we have

$$\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j| \leq \sum_{j=1}^n \|\mathbf{p}\|_\infty = n \|\mathbf{p}\|_\infty$$



Proof: Inequalities between norms (4/4)

Inequalities between norms (4/4).

4. $\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2$: consider the column vectors

$$\mathbf{a} = \begin{pmatrix} |p_1| \\ |p_2| \\ \vdots \\ |p_n| \end{pmatrix} \in \mathbb{R}^n \quad \text{and} \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^n$$

for which $\mathbf{1}' \mathbf{a} = \|\mathbf{p}\|_1$, $\|\mathbf{a}\|_2 = \|\mathbf{p}\|_2$ and $\|\mathbf{1}\|_2 = \sqrt{n}$. By the Cauchy–Schwarz inequality, we get

$$\|\mathbf{p}\|_1 = \mathbf{1}' \mathbf{a} \leq |\mathbf{1}' \mathbf{a}| \leq \|\mathbf{1}\|_2 \|\mathbf{a}\|_2 = \sqrt{n} \|\mathbf{p}\|_2$$



Example: Inequalities between norms

Consider the vector:

$$\mathbf{p} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \in \mathbb{R}^3$$

Its norms are:

$$\|\mathbf{p}\|_{\infty} = \max\{1, 2, 2\} = 2$$

$$\|\mathbf{p}\|_2 = \sqrt{1 + 4 + 4} = 3$$

$$\|\mathbf{p}\|_1 = 1 + 2 + 2 = 5$$

and, with $n = 3$, we have:

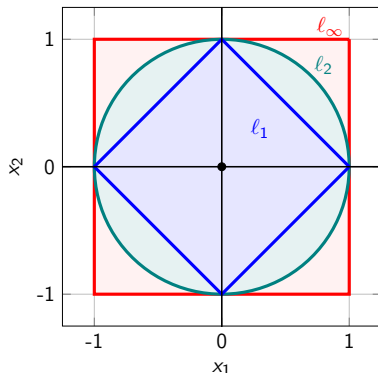
$$\underbrace{2}_{\|\mathbf{p}\|_{\infty}} \leq \underbrace{3}_{\|\mathbf{p}\|_2} \leq \underbrace{5}_{\|\mathbf{p}\|_1} \leq \underbrace{6}_{3\|\mathbf{p}\|_{\infty}}$$

and

$$\underbrace{5}_{\|\mathbf{p}\|_1} \leq \underbrace{3\sqrt{3}}_{\sqrt{3}\|\mathbf{p}\|_2} \approx 5.196$$

Nested unit balls

- In two dimensions, the inequalities $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ mean that the three unit balls are **nested**: the ℓ_1 diamond lies inside the ℓ_2 unit circle, which in turn lies inside the ℓ_∞ square.



Summary

Comparison of the norms:

- ▶ For every $\mathbf{p} \in \mathbb{R}^n$: $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty$
- ▶ Moreover $\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2$ (consequence of Cauchy–Schwarz)
- ▶ Geometrically, in $\mathbb{R}^2$: ℓ_1 diamond $\subseteq \ell_2$ circle $\subseteq \ell_\infty$ square

Summary (1/4)

Norms:

- ▶ A **norm** is a function that assigns a non-negative length to vectors
- ▶ All norms satisfy three properties: non-negativity and definiteness, absolute homogeneity, and triangle inequality

ℓ_1 norm (Manhattan norm):

- ▶ $\|\mathbf{p}\|_1 = \sum_{j=1}^n |p_j|$
- ▶ Sum of absolute values of all components
- ▶ Unit ball: diamond (rotated square) in $\mathbb{R}^2$
- ▶ Used in sparse optimization and robust statistics

Summary (2/4)

ℓ_2 norm (Euclidean norm):

- ▶ $\|\mathbf{p}\|_2 = \sqrt{\sum_{j=1}^n p_j^2}$
- ▶ Represents Euclidean distance from the origin
- ▶ Unit ball: circle in $\mathbb{R}^2$
- ▶ Satisfies Cauchy–Schwarz inequality: $|\mathbf{p}'\mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$
- ▶ Triangle inequality and reverse triangle inequality

Summary (3/4)

Generalized ℓ_2 norm:

- ▶ $\|\mathbf{p}\|_{\mathbf{Q}} = \sqrt{\mathbf{p}'\mathbf{Q}\mathbf{p}}$ where $\mathbf{Q}$ is symmetric positive definite
- ▶ Extends the standard ℓ_2 norm by incorporating a weighting matrix
- ▶ Unit ball: ellipse in $\mathbb{R}^2$ (shape determined by eigenvalues and eigenvectors of $\mathbf{Q}$)
- ▶ Satisfies generalized Cauchy–Schwarz inequality
- ▶ Reduces to standard ℓ_2 norm when $\mathbf{Q} = \mathbf{I}$

Summary (4/4)

ℓ_∞ norm (maximum/Chebyshev norm):

- ▶ $\|\mathbf{p}\|_\infty = \max\{|p_j| : j \in \{1, \dots, n\}\}$
- ▶ Maximum absolute value among all components
- ▶ Unit ball: square in $\mathbb{R}^2$
- ▶ It is half the side length of the smallest axis-aligned hypercube centered at the origin containing the vector

Comparison:

- ▶ $\|\mathbf{p}\|_\infty \leq \|\mathbf{p}\|_2 \leq \|\mathbf{p}\|_1 \leq n \|\mathbf{p}\|_\infty$ and $\|\mathbf{p}\|_1 \leq \sqrt{n} \|\mathbf{p}\|_2$